

Go Backend Plan for Clothing Marketplace (TPRD V2 Aligned)

Stack: Go + Gin + PostgreSQL + Redis + Razorpay + AWS | API: REST + OpenAPI | Architecture: Modular Monolith

What Was Added From TPRD Gap Review

- OTP-based auth (mobile/email), logout/session revocation, and saved size profile APIs.
- Bulk product upload, fabric/performance attribute mapping, and product availability APIs.
- Coupon validation engine, COD eligibility logic, GST + shipping charge calculation.
- Shipment tracking sync, shipping partner webhooks, returns/refunds, COD reconciliation.
- Rule-based fit recommendation engine (height/weight/fit preference -> size + confidence).
- Push notifications + event trigger matrix (order/cart lifecycle events).
- Banner/CMS APIs for marketing/homepage sections.
- Event tracking APIs, conversion funnel analytics, return reason analytics.
- Infra specifics: dev/staging/prod, CDN, ALB/autoscaling, Redis, rollback and DR targets.

Phase-wise Delivery (12-16 Weeks)

Phase 0 (Week 1-2): Platform Foundation

- Repo setup with modules: auth, catalog, inventory, cart, checkout, orders, payments, notifications, cms, fit, analytics.
- Common layers: config, logging, error model, validation, middleware, RBAC support.
- AWS baseline: VPC, ECS, RDS, Redis, S3, ALB, CloudWatch, Secrets Manager.
- Environments and CI/CD: dev, staging, prod with rollback strategy and quality gates.

Phase 1 (Week 2-3): Auth and User Management

- Login/signup via OTP (mobile/email), JWT access/refresh, logout token revocation.
- Password reset (email), role-based access (Admin/Ops/Customer).
- User profile APIs including saved address and saved size profile.

Phase 2 (Week 3-5): Catalog, Discovery, and CMS

- Category/product/SKU CRUD with size-color mapping and availability APIs.
- Fabric and performance attributes; pricing and discount rules.
- Bulk product upload endpoints and job tracking.
- CMS/banner APIs for hero, featured, and campaign sections.

Phase 3 (Week 5-8): Cart, Checkout, Orders, Payments

- Cart create/update/remove APIs, coupon validation, shipping charges and GST computation.
- COD eligibility checks by pincode/order value and payment method selection support.
- Razorpay payment intent/order creation, webhook verification, idempotent capture flow.
- OMS lifecycle + shipment tracking sync + returns/refunds + COD reconciliation.

Phase 4 (Week 8-10): Inventory and Admin Operations

- Real-time SKU stock APIs, checkout-time stock lock/reservation, low-stock alerts.
- Manual stock adjustment APIs and restock recommendation inputs.
- Admin APIs for product/inventory/orders/users/CMS with audit logs.

Phase 5 (Week 10-12): Fit, Notifications, and Analytics

- Fit recommendation engine (rules-based) returning recommended size and confidence score.
- Email/SMS/Push notifications for order/cart lifecycle events.
- Event ingestion APIs, server-side analytics, conversion funnel, SKU performance, return reasons.

Phase 6 (Week 12-16): Hardening and Go-Live

- Security: SSL, WAF/firewall, rate limiting, secrets rotation.
- Reliability: health checks, monitoring, error tracking, alerting and incident runbooks.
- Performance: DB tuning, Redis cache strategy, autoscaling thresholds, load testing.
- Backup/DR: retention policy, restore tests, RTO/RPO sign-off and production cutover.

API and Interface Requirements

- REST under `/api/v1` with OpenAPI contracts for all customer/admin endpoints.
- Core interfaces: `PaymentProvider`, `ShippingProvider`, `OtpProvider`, `NotificationProvider`, `TaxCalculator`.

- Idempotency required for checkout/order/payment webhooks and retries.

Acceptance Criteria

- Successful purchase flow: browse -> cart -> coupon -> checkout -> Razorpay payment -> confirmation.
- Failed payment retry does not create duplicate orders/payments.
- Inventory reservation prevents overselling under concurrency.
- Shipment webhook updates order timeline correctly.
- Returns/refunds/COD reconciliation reports are accurate.
- Analytics dashboards reconcile with transactional source tables.

Assumptions

- India-first launch (INR, GST, Razorpay, COD support).
- Single brand initially, schema remains multi-brand ready.
- Web + mobile clients consume the same backend APIs.